# Modeling Dopamine and Serotonin in Reinforcement Learning: A Computational Psychiatry Approach to Depression and ADHD

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## Introduction

Psychiatric disorders like depression and ADHD are typically described in terms of symptoms and neurochemical abnormalities, but this framing lacks explanatory depth: it says what changes, not how those changes alter learning, decision-making, and motivation. Computational psychiatry addresses this by linking neural mechanisms to behavior through formal models that generate testable hypotheses (Lan & Browning, 2022). Reinforcement learning (RL) is especially well-suited to this effort because it decomposes behavior into separable components—prediction error, learning rate, expected value, action policy—that can vary independently (Chen et al., 2015).

This decomposition becomes powerful when combined with what is known about dopamine and serotonin. A meta-analysis of 107 placebo-controlled pharmacology studies found that dopaminergic manipulations were linked to reward learning and sensitivity (SMD = 0.26) and reward vigor (SMD = 0.32), whereas serotonergic manipulations were selectively associated with punishment learning (SMD = 0.32) and aversive Pavlovian processes (SMD = 0.36), with no meaningful association with reward learning (SMD = −0.29) (Mkrtchian et al., 2025). This double dissociation motivates our modeling approach.

We implemented a Q-learning model with separate parameters for dopamine-related reward updating and serotonin-related aversive processing, then tested how altering these parameters reproduces behavioral phenotypes associated with depression and ADHD. Our goal is to generate mechanistic, hypothesis-level predictions, not clinical recommendations.

## Background

### *Reinforcement Learning*

In RL, an agent learns by comparing expected and received outcomes. The prediction error (δ) is the core teaching signal:

$$\delta_t = r_t - Q(a_t)$$

where $r_t$ is the received reward and $Q(a_t)$ is the current value estimate for action $a_t$. Values update as:

$$Q_{t+1}(a_t) = Q_t(a_t) + \alpha \cdot \delta_t$$

with α as the learning rate. Actions are selected via softmax with inverse temperature β: higher β yields more deterministic exploitation of learned values; lower β produces stochastic exploration. This framework is valuable in psychiatry precisely because it makes mechanisms explicit—depression can reflect blunted reward sensitivity (low $\alpha^+$ or reward weighting) rather than a vague motivational deficit, and ADHD can reflect inconsistent value-to-action translation (low β) rather than an inability to learn (Chen et al., 2015).

### *Dopamine*

Dopamine neurons encode reward prediction errors: they fire for unexpected rewards, shift responses to reward-predicting cues after learning, and pause when expected rewards are omitted (Schultz, 1998). Glimcher (2011) synthesized this into the dopamine reward prediction error hypothesis, in which phasic dopamine drives synaptic modification across the frontal cortex and basal ganglia. Pharmacological evidence confirms this reward-selectivity in humans (Mkrtchian et al., 2025), justifying our treatment of dopamine as a gain parameter on positive prediction error updating.

### *Serotonin*

Serotonin's computational role is less settled. Michely et al. (2020) found that one week of SSRI administration enhanced punishment learning and reduced reward learning—consistent with serotonin as a punishment-sensitivity modulator. However, Colwell et al. (2023), using a more direct serotonergic manipulation (fenfluramine), found that elevated serotonin instead reduced aversive outcome sensitivity

while increasing behavioral inhibition. These accounts are not equivalent: one predicts stronger negative updating; the other predicts more cautious, deterministic choice. We test both in our model.

## Competing Computational Roles of Serotonin

We formalize the two serotonin hypotheses as follows. Under Model A (Punishment Sensitivity), serotonin scales $\alpha^-$, the learning rate for negative prediction errors. Higher $\alpha^-$ means the agent updates values more strongly after worse-than-expected outcomes, predicting more post-loss switching and stronger aversive conditioning. This operationalizes the SSRI findings of Michely et al. (2020) and the meta-analytic punishment-learning effect of Mkrtchian et al. (2025).

Under Model B (Behavioral Inhibition), serotonin scales $\beta$, the softmax inverse temperature. Higher $\beta$ makes the action policy more deterministic—the agent commits more firmly to its current best estimate without changing how outcomes are learned from. This operationalizes the Colwell et al. (2023) finding that serotonin increases impulse control and reduces response to aversive outcomes without amplifying punishment learning per se.

The distinction matters for psychiatric interpretation. The punishment-sensitivity account predicts valence-asymmetric updating; the inhibition account predicts globally more cautious behavior. In a reversal learning task, these produce different switch rate and recovery profiles, allowing the task to partially discriminate between them.

## Model and Task Design

### *Probabilistic Reversal Learning Task*

We used a two-armed probabilistic reversal learning task across 160 trials. One arm delivered reward with $p = 0.75$ (the 'good' arm); the other with $p = 0.25$ (the 'bad' arm). At trials 40, 80, and 120, contingencies reversed. This task is sensitive to both learning rates (how quickly the agent detects the reversal) and $\beta$ (whether learned values translate into consistent choices). It also mirrors the paradigm used in the ADHD empirical literature we draw on (Aster et al., 2024).

### *Q-Learning with Asymmetric Updating*

We used Q-learning with separate learning rates for positive and negative prediction errors:

$$\delta_t = r_t - Q(a_t)$$

$$Q_{t+1}(a_t) = Q_t(a_t) + \alpha^+ \max(\delta_t, 0) + \alpha^- \min(\delta_t, 0)$$

Actions were selected by softmax:

$$P(a_t = a) = \exp(\beta \cdot Q(a)) / \Sigma_j \exp(\beta \cdot Q(j))$$

All Q-values were initialized at 0.5. The three free parameters—$\alpha^+$, $\alpha^-$, $\beta$—map onto neuromodulatory systems as follows: k_DA scales $\alpha^+$ (dopamine gain on reward learning); k_5HT scales either $\alpha^-$ (Model A) or $\beta$ (Model B). 200 independent agents with different random seeds were run per condition.

### *Parameter Conditions*

| Condition | $\alpha^+$ | $\alpha^-$ | $\beta$ | Rationale |
|---|---|---|---|---|
| Baseline | 0.30 | 0.30 | 5.0 | Standard agent |
| High Dopamine | 0.60 | 0.30 | 5.0 | ↑ reward learning ($\alpha^+$) |
| High 5HT – Model A | 0.30 | 0.60 | 5.0 | ↑ punishment sensitivity ($\alpha^-$) |
| High 5HT – Model B | 0.30 | 0.30 | 9.0 | ↑ behavioral inhibition ($\beta$) |
| Depression-like | 0.10 | 0.30 | 3.0 | ↓ $\alpha^+$, ↓ $\beta$ (blunted reward) |
| ADHD-like | 0.30 | 0.30 | 1.5 | ↓ $\beta$ (noisy exploitation) |

The depression-like set was motivated by Chen et al. (2015), who reviewed evidence that depression is primarily associated with reduced reward sensitivity rather than altered punishment learning. We accordingly reduced $\alpha^+$ and $\beta$, leaving $\alpha^-$ at baseline. The ADHD-like set was motivated by Aster et al. (2024) and Addicott et al. (2021), both of which support a model of ADHD as impaired value exploitation rather than impaired value learning, operationalized as low $\beta$.

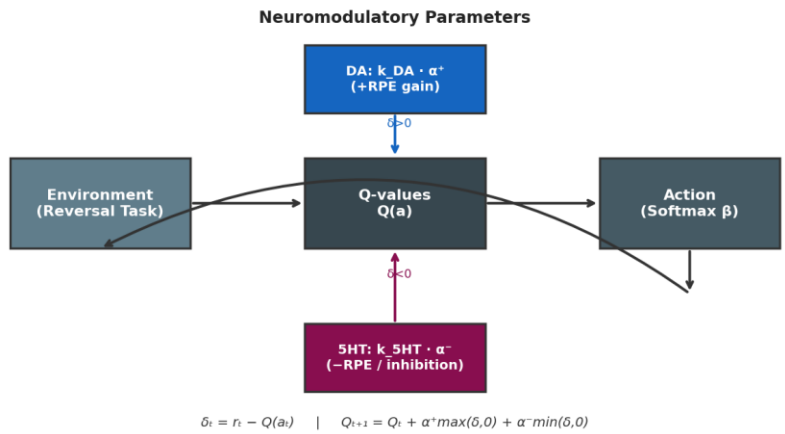


*Figure 1. Model schematic. Dopamine (blue) scales $\alpha^+$ (positive RPE); serotonin scales either $\alpha^-$ (Model A, red) or $\beta$ (Model B). Softmax translates Q-values into action probabilities.*

## Simulation Results

We report three metrics: mean reward across all 160 trials, trial-to-trial switch rate, and post-reversal recovery (mean reward in the 15 trials after each reversal). Table 2 summarizes results; Figures 2–4 show learning curves, behavioral phenotypes, and Q-value trajectories.

| Condition | Mean Reward | Switch Rate | Post-Rev. Recovery |
|---|---|---|---|
| Baseline | 0.632 | 0.218 | 0.526 |
| High Dopamine | 0.635 | 0.184 | 0.534 |
| High 5HT – Model A | 0.635 | 0.235 | 0.569 |
| High 5HT – Model B | 0.650 | 0.117 | 0.514 |
| Depression-like | 0.569 | 0.410 | 0.487 |
| ADHD-like | 0.560 | 0.438 | 0.517 |

*Table 2. Simulation summary statistics across 200 agents per condition.*

### *Dopamine and Serotonin Manipulations*

High dopamine ($\alpha^+ = 0.60$) produced a lower switch rate (0.184 vs. 0.218 baseline), reflecting faster Q-value differentiation: stronger positive prediction error updates cause the better arm's value to separate from the worse arms more quickly, and the softmax policy exploits that difference more consistently. This aligns with the interpretation that dopamine facilitates stable reward pursuit (Glimcher, 2011).

The two serotonin conditions diverged as predicted. High 5HT Model A ($\alpha^- = 0.60$) produced a higher switch rate (0.235) and notably better post-reversal recovery (0.569), because losses after a reversal

rapidly deflate the now-bad arm's Q-value, accelerating the switch to the now-good arm. High 5HT Model B ($\beta = 9.0$) produced the lowest switch rate of all conditions (0.117) and the highest overall mean reward (0.650), because deterministic exploitation reduces random errors in stable periods—but post-reversal recovery was slightly lower (0.514) since a high-$\beta$ policy is slow to abandon an arm whose value has only partially declined.

### *Psychiatric Phenotypes*

The depression-like condition ($\alpha^+ = 0.10$, $\beta = 3.0$) produced mean reward = 0.569 and switch rate = 0.410—the lowest reward and second-highest switching. Crucially, Figure 4 shows that Q-values differentiate slowly and weakly across the session, even during stable pre-reversal periods. Reward information simply does not drive strong value separation, so behavior remains near-random. This matches the computational psychiatry characterization of depression as blunted reward prediction error signaling and reduced incentive salience, not just sadness (Chen et al., 2015).

The ADHD-like condition ($\beta = 1.5$) produced mean reward = 0.560 and the highest switch rate (0.438). Strikingly, Figure 4 shows that Q-values do track reversals—the model learns—but the low $\beta$ means the softmax policy does not translate that learning into consistent choices. This distinguishes ADHD from depression mechanistically: impaired value expression versus impaired value learning. Both produce poor behavioral outcomes, but for different reasons, a distinction that surface behavioral data alone would obscure (Addicott et al., 2021; Aster et al., 2024).

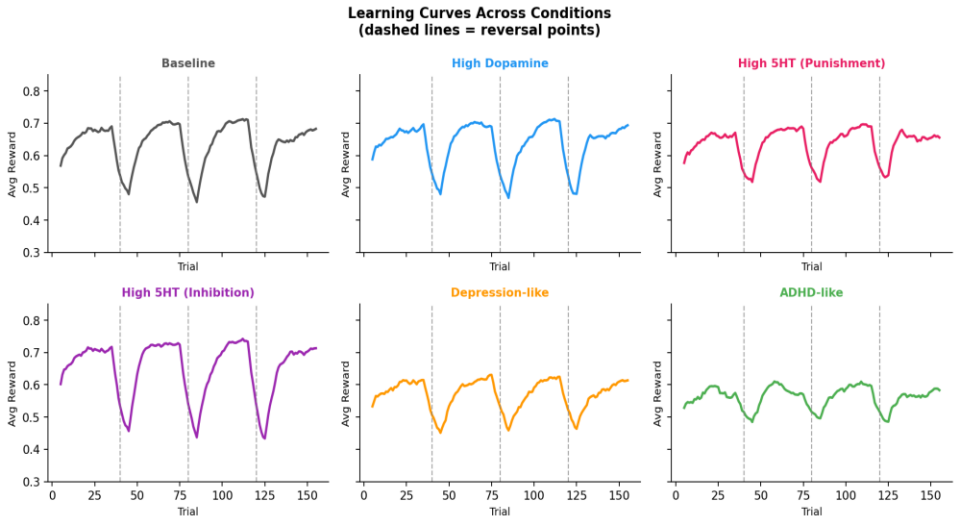


*Figure 2. Smoothed reward curves (10-trial window) across conditions. Dashed lines = reversals. Depression-like and ADHD-like show flat, low trajectories; High 5HT Model B shows highest stable-period reward.*

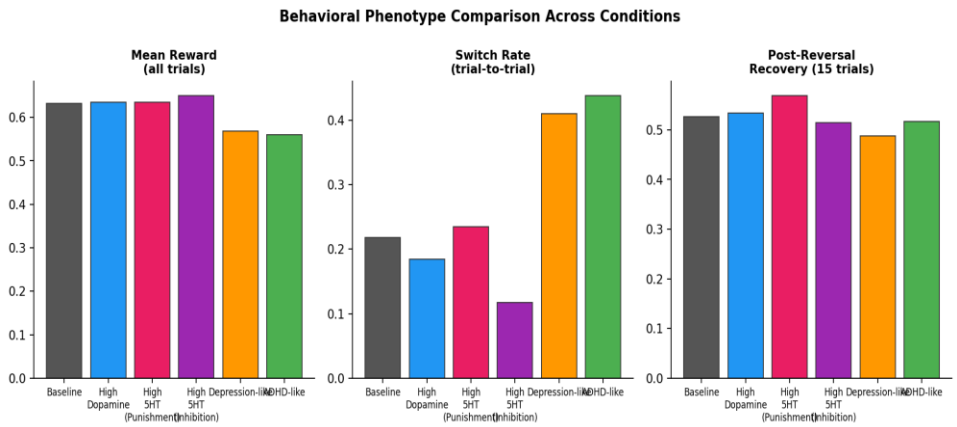


*Figure 3. Behavioral phenotype comparison: mean reward, switch rate, and post-reversal recovery. The two serotonin models diverge on switch rate and recovery, consistent with their distinct mechanisms.*

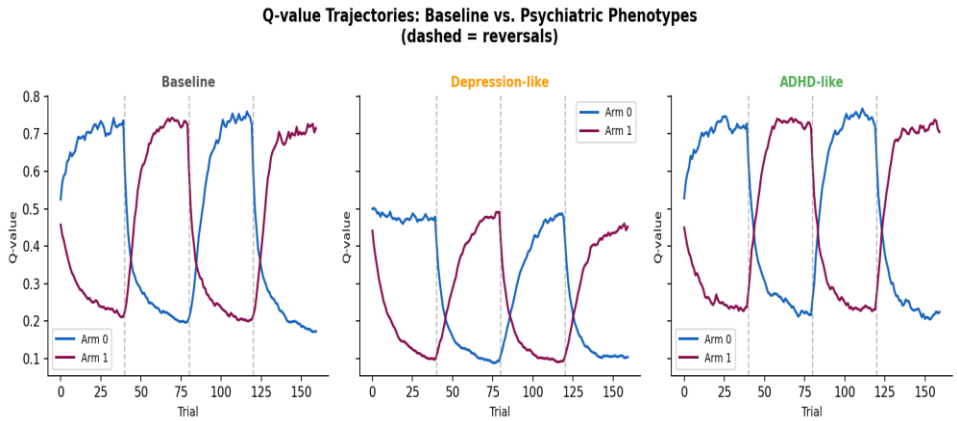


*Figure 4. Q-value trajectories (averaged across 200 agents) for baseline, depression-like, and ADHD-like conditions. Depression shows slow, weak differentiation; ADHD shows tracking without exploitation.*

## Depression and ADHD Interpretation

The model's value lies not in reducing these disorders to two or three numbers, but in making the mechanisms of behavioral change explicit and separable. For depression, the clearest computational signature is blunted reward processing: reduced prediction error signaling, lower expected value, and decreased reward sensitivity (Chen et al., 2015). In the model, even when rewards are available, they do not drive meaningful value updates—the agent does not shift behavior toward them. This maps onto the clinical picture of anhedonia and reduced motivation, framed not as sadness but as a failure of reward to guide action.

ADHD presents a different computational profile. The problem is not that reward fails to be learned, but that learned values are not consistently translated into behavior. Aster et al. (2024) showed that ADHD patients switch choices more randomly with reduced reinforcement sensitivity; Addicott et al. (2021) linked ADHD to excessive exploration on a multi-armed bandit. Both are consistent with a low-β characterization. The model shows why these disorders can look similar at the surface (both produce low reward and high switching) while having distinct mechanisms.

Regarding serotonergic medications: the model cautions against treating SSRIs as simple mood elevators. Serotonergic enhancement shifts how outcome valence shapes learning—either amplifying punishment updating (Model A) or increasing behavioral inhibition (Model B). Neither maps simply onto 'feeling better,' and the two make different predictions about pharmacological effects on reversal learning that could in principle be tested empirically (Lan & Browning, 2022; Michely et al., 2020).

## Discussion

Our results support two main claims. First, the two serotonin formalizations produce behaviorally distinct signatures in reversal learning—diverging on switch rate and post-reversal recovery—suggesting the empirical distinction between punishment sensitivity and behavioral inhibition has real computational traction. Second, depression-like and ADHD-like parameter configurations produce different mechanisms

of behavioral impairment despite similar surface performance, a dissociation that formal modeling makes visible.

We interpret these results cautiously. Lan & Browning (2022) emphasize that the evidence directly linking RL parameter changes to antidepressant outcomes in clinical populations remains limited. Our simulations generate predictions, not clinical accounts. The model does not claim to explain the etiology of depression or ADHD—it demonstrates that a minimal computational framework can generate qualitatively consistent behavioral phenotypes, which may help structure hypotheses about mechanisms and treatments.

## Limitations and Future Directions

Several limitations are important to note. First, we treated dopamine and serotonin as independent scalars, whereas Diederen & Fletcher (2021) argue they interact in complementary ways within model-based learning circuits. Second, our single two-armed task may not generalize to the richer contexts studied empirically (e.g., six-armed bandits, volatile environments). Third, mechanistic basal-ganglia models of ADHD (Stringer et al., 2016) are considerably more detailed, accounting for tonic/phasic dopamine distinctions and circuit-level dynamics our model cannot represent.

Future work could fit the model to actual choice data from depressed or ADHD participants to assess parameter recovery and individual differences. Adding model-based planning components, as suggested by Diederen & Fletcher (2021), would allow testing whether serotonin's role in sustaining future reward expectations can be distinguished from its role in immediate feedback processing.

## Conclusion

We implemented a Q-learning model with asymmetric learning rates and a softmax policy to examine how dopamine- and serotonin-related parameters shape behavior in a probabilistic reversal learning task. We found that the two candidate serotonin accounts—punishment sensitivity and behavioral inhibition—produce discriminable behavioral signatures, and that depression-like and ADHD-like parameter configurations produce qualitatively distinct mechanisms of impaired performance despite

similar outcomes on the surface. These results illustrate the core promise of computational psychiatry: formal models can decompose symptom-level descriptions into testable mechanistic claims, potentially connecting pharmacology, neural systems, and behavior in a principled way.

## References


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## Appendix: Simulation Code

*Full code is provided as simulation_code.py (submitted separately). Key excerpts shown below.*

```
import numpy as np
import matplotlib.pyplot as plt

N_TRIALS = 160;  N_AGENTS = 200;  REVERSAL_AT = [40, 80, 120]
P_WIN_GOOD = 0.75;  P_WIN_BAD = 0.25

def run_agent(alpha_pos, alpha_neg, beta, n_trials=160, n_agents=200):
    """Q-learning with asymmetric learning rates and softmax."""
    all_rewards  = np.zeros((n_agents, n_trials))
    all_switches = np.zeros((n_agents, n_trials - 1), dtype=int)
    all_q        = np.zeros((n_agents, n_trials, 2))
    for ag in range(n_agents):
```

```
        Q = np.array([0.5, 0.5])
        prev = None
        for t in range(n_trials):
            ev = np.exp(beta * Q)
            p  = ev / ev.sum()                    # softmax
            c  = np.random.choice(2, p=p)         # action
            r  = task_reward(t, c, ag)            # outcome
            d  = r - Q[c]                         # prediction error
            if d >= 0:  Q[c] += alpha_pos * d   # positive RPE
            else:       Q[c] += alpha_neg * d   # negative RPE
            all_rewards[ag, t]  = r
            all_q[ag, t]        = Q.copy()
            if prev is not None:
                all_switches[ag, t-1] = int(c != prev)
            prev = c
    return all_rewards, all_q, all_switches

# Parameter conditions
PARAMS = {
    "Baseline":             dict(alpha_pos=0.30, alpha_neg=0.30, beta=5.0),
    "High Dopamine":        dict(alpha_pos=0.60, alpha_neg=0.30, beta=5.0),
    "High 5HT (Model A)":   dict(alpha_pos=0.30, alpha_neg=0.60, beta=5.0),
    "High 5HT (Model B)":   dict(alpha_pos=0.30, alpha_neg=0.30, beta=9.0),
    "Depression-like":      dict(alpha_pos=0.10, alpha_neg=0.30, beta=3.0),
    "ADHD-like":            dict(alpha_pos=0.30, alpha_neg=0.30, beta=1.5),
}
```

## AI Usage

Claude (Anthropic, claude-sonnet-4-6) was used in this project as follows. We used it to (1) help identify and organize the relevant literature, summarizing which papers were most relevant to each section; (2) suggest the paper structure and section organization; (3) provide feedback on drafts of individual sections and suggest edits for clarity; and (4) generate the Python simulation code and figures based on the model specification we designed. All scientific claims, parameter choices, and interpretations were verified against the primary literature by the group. The final writing was produced by the group members, with AI assistance used to refine phrasing and transitions. We take responsibility for accuracy.